

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 3

Date Due: See canvas

This assignment has a total of **(250 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 3.1 (20 pts) Albino rats used to study the hormonal regulation of a metabolic pathway are injected with a drug that inhibits body synthesis of protein. The probability that a rat will die from the drug before the experiment is over is 0.2.

- (i) (10 pts) If 10 animals are treated with the drug, how many are expected to die before the experiment ends?
- (ii) (10 pts) What is the probability that at least eight will survive? Would you be surprised if at least five died during the course of the experiment? Explain, based on the probability of this occurring.

Exercise 3.2 (20 pts) It has been found that 80% of all printers used on home computers operate correctly at the time of installation. The rest require some adjustment. A particular dealer sells 10 units during a given month.

- (i) (10 pts) Find the probability that at least nine of the printers operate correctly upon installation.
- (ii) (10 pts) Consider 5 months in which 10 units are sold per month. What is the probability that at least 9 units operate correctly in each of the 5 months?

Exercise 3.3 (10 pts) A company is manufacturing highway emergency flares. Such flares are supposed to burn for an average of 20 minutes. Every hour a sample of flares is collected, and their average burn time is determined. If the manufacturing process is working correctly, there is a 68% chance that the average burn time of the sample will be between 14 minutes and 26 minutes. The quality engineer in charge of the process believes that if 4 of 5 samples fall outside these bounds then this is a signal that the process might not be performing as expected. Each morning the sampling begins anew. Let X denote the number of samples drawn in order to obtain the fourth sample whose average value is outside of the above bounds. Find the probability that for a given morning $X = 5$ and hence there seems to be a problem right away.

Exercise 3.4 (40 pts) Twenty microprocessor chips are in stock. Three have etching errors that cannot be detected by the naked eye. Five chips are selected and installed in field equipment.

- (i) (10 pts) Find the density for X , the number of chips selected that have etching errors.
- (ii) (10 pts) Find $\mathbb{E}[X]$ and $\text{Var } X$.
- (iii) (10 pts) Find the probability that no chips with etching errors will be selected.
- (iv) (10 pts) Find the probability that at least one chip with an etching error will be chosen.

Exercise 3.5 (20 pts) *Escherichia coli*, a bacterium often found in the human digestive tract, can mutate from being streptomycin sensitive to being streptomycin resistant, which can cause the individual involved to become resistant to the antibiotic streptomycin. Assume that there is an average of two streptomycin-resistant bacteria on cultures drawn from a particular patient. Each culture has an area of 80 square centimeters.

- (i) (10 pts) What is the probability that a one-square-centimeter random sample from a single culture will contain at least one resistant bacterium?
- (ii) (10 pts) What is the probability that at least one will be found in 5 randomly selected one-square-centimeter samples? (Assume that the 5 samples are independent of one another.)

Exercise 3.6 (20 pts) Given proposed cumulative distributions as below. In each case, find the “density” that would be associated with each, and decide whether it really does define a valid continuous density. If it does not, explain what property fails.

- (i) (10 pts)

$$F(x) = \begin{cases} 0, & x < -1 \\ x + 1, & -1 \leq x \leq 0 \\ 1, & x > 0 \end{cases}$$

(ii) (10 pts)

$$F(x) = \begin{cases} 0, & x \leq 0 \\ x^2, & 0 < x \leq \frac{1}{2} \\ \frac{x}{2}, & \frac{1}{2} < x \leq 1 \\ 1, & x > 1 \end{cases}$$

Exercise 3.7 (40 pts) Let X denote the amount of time in hours that a battery on a solar calculator will operate adequately between exposures to light sufficient to recharge the battery. Assume that the density for X is given by

$$f(x) = \frac{50}{6x^3}, \quad 2 < x < 10$$

- (i) (10 pts) Verify that this is a valid continuous density.
- (ii) (10 pts) Find the expression for the cumulative distribution function for X , and use it to find the probability that a randomly selected solar battery will last at most 4 hours before needing to be recharged.
- (iii) (10 pts) Find the average time that a battery will last before needing to be recharged.
- (iv) (10 pts) Find $\mathbb{E}[X^2]$, and use this to find $\text{Var } X$.

Exercise 3.8 (10 pts) California is hit every year by approximately 500 earthquakes that are large enough to be felt. However, those of destructive magnitude occur, on the average, once a year. Find the probability that at least 3 months elapse before the first earthquake of destructive magnitude occurs.

Exercise 3.9 (40 pts) Most galaxies take the form of a flattened disc, with the major part of the light coming from this very thin fundamental plane. The degree of flattening differs from galaxy to galaxy. In the Milky Way Galaxy most gases are concentrated near the center of the fundamental plane. Let X denote the perpendicular distance from this center to a gaseous mass. X is normally distributed with mean 0 and standard deviation 100 parsecs. (A parsec is equal to approximately 19.2 trillion miles.)

- (i) (10 pts) Sketch a graph of the density for X . Indicate on this graph the probability that a gaseous mass is located within 200 parsecs of the center of the fundamental plane. Find this probability.
- (ii) (10 pts) Approximately what percentage of the gaseous masses are located more than 250 parsecs from the center of the plane?
- (iii) (10 pts) What distance has the property that 20% of the gaseous masses are at least this far from the fundamental plane?
- (iv) (10 pts) What is the moment generating function for X ?

Exercise 3.10 (30 pts) (*Log-normal distribution.*) The log-normal distribution is the distribution of a random variable whose natural logarithm follows a normal distribution. Thus if X is a normal random variable, then $Y = e^X$ follows a log-normal distribution. Complete the argument below, thus deriving the density for a log-normal random variable.

Let X be normal with mean μ and variance σ^2 . Let G denote the cumulative distribution function for $Y = e^X$, and let F denote the cumulative distribution function for X .

- (i) (10 pts) Show that $G(y) = F(\ln y)$.
- (ii) (10 pts) Show that $G'(y) = F'(\ln y)/y$.
- (iii) (10 pts) Show that the density for Y is given by

$$g(y) = \frac{1}{\sqrt{2\pi}\sigma y} \exp\left[-\frac{(\ln y - \mu)^2}{2\sigma^2}\right], \quad y > 0, \quad -\infty < \mu < \infty, \sigma > 0$$

Note that μ and σ are the mean and standard deviation of the underlying normal distribution; they are not the mean and standard deviation of Y itself.